

Table 1: Complete performance comparison at 100k steps vs SafePO at 5M steps.

	Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
	SafePO	1c3s5z	7.64 ± 0.16	9.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	1c3s5z	19.29 ± 0.14	4.23 ± 0.27	0.82 ± 0.02	100k	0
	SafePO	1c3s5z	7.79 ± 0.05	9.00 ± 0.00	0.00 ± 0.00	5M	4
	Safe Dreamers	1c3s5z	19.15 ± 0.53	4.83 ± 0.94	0.79 ± 0.12	100k	4
	SafePO	2m_vs_1z	2.71 ± 0.18	2.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	2m_vs_1z	19.30 ± 0.49	0.14 ± 0.08	0.95 ± 0.04	100k	0
	SafePO	2m_vs_1z	3.13 ± 0.01	2.00 ± 0.00	0.00 ± 0.00	5M	1
	Safe Dreamers	2m_vs_1z	19.34 ± 0.38	0.14 ± 0.09	0.95 ± 0.03	100k	1
	SafePO	2s3z	3.69 ± 0.02	5.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	2s3z	18.44 ± 0.25	2.28 ± 0.38	0.77 ± 0.04	100k	0
	SafePO	2s3z	4.86 ± 0.04	5.00 ± 0.00	0.00 ± 0.00	5M	4
	Safe Dreamers	2s3z	17.37 ± 1.38	2.98 ± 0.64	0.63 ± 0.15	100k	4
	SafePO	2s_vs_1sc	2.44 ± 0.10	2.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	2s_vs_1sc	19.95 ± 0.21	0.23 ± 0.11	0.96 ± 0.03	100k	0
	SafePO	2s_vs_1sc	4.21 ± 0.21	2.00 ± 0.00	0.00 ± 0.00	5M	1
	Safe Dreamers	2s_vs_1sc	19.57 ± 0.49	0.31 ± 0.10	0.92 ± 0.06	100k	1
	SafePO	3m	2.06 ± 0.10	3.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	3m	19.57 ± 0.25	0.35 ± 0.14	0.97 ± 0.02	100k	0
	SafePO	3m	2.55 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	5M	1
	Safe Dreamers	3m	19.46 ± 0.42	0.60 ± 0.17	0.96 ± 0.03	100k	1
	SafePO	3s5z_vs_3s6z	3.47 ± 0.02	8.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	3s5z_vs_3s6z	9.60 ± 0.84	8.00 ± 0.00	0.00 ± 0.00	100k	0
	SafePO	3s5z_vs_3s6z	3.86 ± 0.04	8.00 ± 0.00	0.00 ± 0.00	5M	4
	Safe Dreamers	3s5z_vs_3s6z	13.28 ± 1.10	7.54 ± 0.40	0.11 ± 0.06	100k	4
	SafePO	3s_vs_3z	6.04 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	3s_vs_3z	19.83 ± 0.20	0.95 ± 0.18	0.96 ± 0.02	100k	0
	SafePO	3s_vs_3z	6.28 ± 0.10	3.00 ± 0.00	0.00 ± 0.00	5M	1
	Safe Dreamers	3s_vs_3z	19.93 ± 0.20	0.79 ± 0.48	0.96 ± 0.04	100k	1
	SafePO	3s_vs_4z	4.39 ± 0.03	3.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	3s_vs_4z	18.19 ± 1.57	2.64 ± 0.34	0.23 ± 0.16	100k	0
	SafePO	3s_vs_4z	4.55 ± 0.01	3.00 ± 0.00	0.00 ± 0.00	5M	1
	Safe Dreamers	3s_vs_4z	16.96 ± 0.74	2.47 ± 0.35	0.31 ± 0.11	100k	1
	SafePO	3s_vs_5z	3.44 ± 0.03	3.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	3s_vs_5z	16.88 ± 1.58	2.67 ± 0.40	0.15 ± 0.19	100k	0
	SafePO	3s_vs_5z	3.52 ± 0.02	3.00 ± 0.00	0.00 ± 0.00	5M	1
	Safe Dreamers	3s_vs_5z	17.44 ± 1.49	2.67 ± 0.43	0.17 ± 0.22	100k	1
	SafePO	8m	1.43 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	8m	18.64 ± 2.02	2.81 ± 1.91	0.86 ± 0.21	100k	0
	SafePO	8m	1.63 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	5M	4
	Safe Dreamers	8m	17.26 ± 1.04	5.03 ± 0.42	0.69 ± 0.10	100k	4
	SafePO	MMM	1.87 ± 0.01	10.00 ± 0.00	0.00 ± 0.00	5M	0
	Safe Dreamers	MMM	17.05 ± 2.36	6.77 ± 1.79	0.59 ± 0.25	100k	0
	SafePO	MMM	1.89 ± 0.04	10.00 ± 0.00	0.00 ± 0.00	5M	4
	Safe Dreamers	MMM	17.02 ± 1.32	7.88 ± 0.16	0.55 ± 0.07	100k	4

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Table 1 – continued

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
SafePO	bane_vs_bane	19.26 ± 0.01	22.55 ± 0.02	0.79 ± 0.01	5M	0
Safe Dreamers	bane_vs_bane	17.75 ± 2.82	13.91 ± 7.11	0.62 ± 0.39	100k	0
SafePO	bane_vs_bane	19.37 ± 0.09	22.40 ± 0.06	0.80 ± 0.02	5M	4
Safe Dreamers	bane_vs_bane	19.91 ± 0.11	6.37 ± 1.20	0.98 ± 0.02	100k	4